

Two lenses are combined to make a microscope with a combined lens strength of 30 D. If one lens has a strength of 10 D, what is the strength of the second lens?

- ☐ A. 3 D [44%]
- ☐ B. 5 D [3%]
- ☐ C. 10 D [2%]
- ✓ ☒ D. 20 D [49%]

Correct

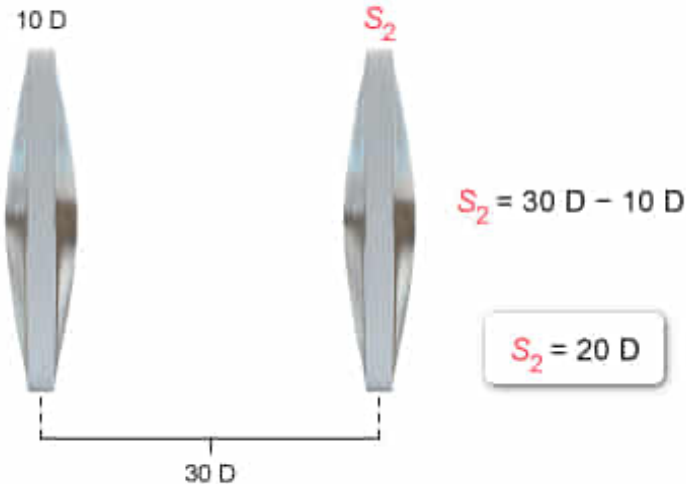
50% answered correctly

Explanation:

Combined strength of two lenses

$S = S_1 + S_2$

- S Overall lens strength
- S_1 Strength of first lens
- S_2 Strength of second lens



$S_2 = 30 \text{ D} - 10 \text{ D}$

$S_2 = 20 \text{ D}$

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The **strength** S of a lens is given in units of **diopters** (D), which is the inverse of meters. The lens strength equals the inverse of the **focal length** f of the lens:

$$S = \frac{1}{f}$$

Lenses can be **combined** to achieve different values of S . Combining

two lenses with strengths S_1 and S_2 yields an overall S equal to the sum of S_1 and S_2 :

$$S = S_1 + S_2$$

In this question, an overall S of 30 D is needed for the microscope and S_1 equals 10 D. Substituting these values into the above equation gives:

$$30 \text{ D} = 10 \text{ D} + S_2$$

Solving for S_2 yields:

$$S_2 = 30 \text{ D} - 10 \text{ D} = 20 \text{ D}$$

Therefore, the strength of the second lens is 20 D.

(Choice A) 3 D results from the ratio of the strengths of the combination of the two lenses and the 10 D lens.

(Choice B) A 5 D lens in this combination produces an overall strength of 15 D, not 30 D.

(Choice C) 10 D is half the lens strength needed for an overall strength of 30 D.

Educational objective:

The combined lens strength of a two-lens system equals the sum of the strengths of each lens.

Time spent:0 sec

Subject:

Physics

QID:404154

System:

Light and Sound